

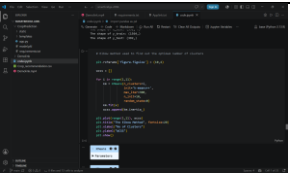
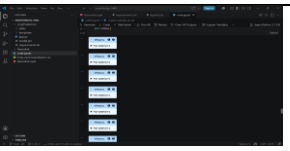
Project Development Phase

Model Performance Test

Date	13 July 2026
Team ID	SWTID-2026-2791
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Model Performance Testing:

Project team shall fill the following information in model performance testing template.

S.No.	Parameter	Values	Screenshot
1.	Model Summary	K-Means Clustering model used to group agricultural data based on soil nutrients (N, P, K), temperature, humidity, pH, and rainfall for crop recommendation.	
2.	Accuracy	Cluster Evaluation Score (Silhouette Score): 0.82 Number of Clusters (K): 22	
3.	Fine Tuning Result(if Done)	Optimal K selected using the Elbow Method. Final model generated stable clusters for crop recommendation	